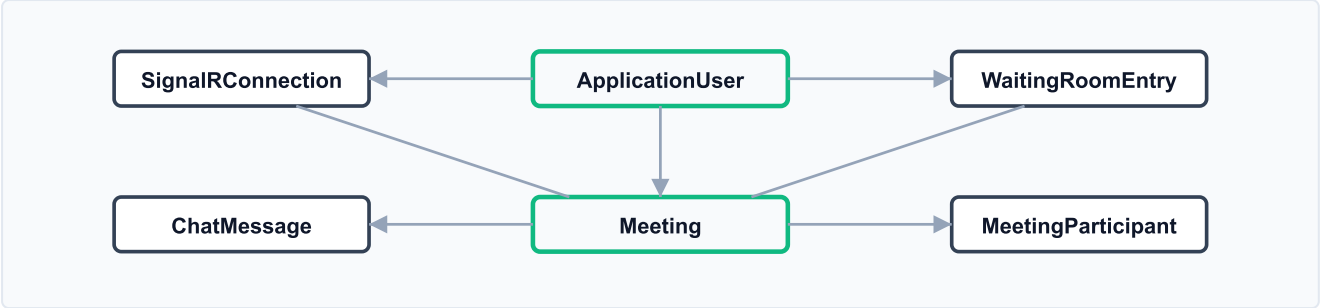
High-Level Topological Diagram



Core Domain Entities

ApplicationUser	Identity Schema
PK Id	nvarchar(450)
FullName	nvarchar(100)
MilitaryRank	nvarchar(50)
Email	nvarchar(256)
PasswordHash	nvarchar(max)

Meeting	dbo
PK Id	uniqueidentifier
FK HostId	nvarchar(450)
Title	nvarchar(200)
LiveKitRoomName	nvarchar(100)
Status	nvarchar(20)
MinIORecordingUrl	nvarchar(500)

MeetingParticipant	dbo
PK Id	uniqueidentifier
FK MeetingId	uniqueidentifier
FK UserId	nvarchar(450)
JoinedAt	datetime2
Role	nvarchar(50)

Persistent SignalR Entities

These tables map transient WebSocket and meeting states into strict MSSQL persistence, protecting the application from memory loss during server restarts.

SignalRConnection	dbo
PK ConnectionId	nvarchar(100)
FK UserId	nvarchar(450)
FK MeetingId	uniqueidentifier null
ConnectedAt	datetime2
Device	nvarchar(100)

ChatMessage	dbo
PK Id	uniqueidentifier
FK MeetingId	uniqueidentifier
FK SenderId	nvarchar(450)
MessageText	nvarchar(max)
SentAt	datetime2

WaitingRoomEntry	dbo
PK Id	uniqueidentifier
FK MeetingId	uniqueidentifier
FK UserId	nvarchar(450)
Status	nvarchar(20)
RequestedAt	datetime2

SignalR Persistence Lifecycle

When a client connects to the Hub, a new record is inserted into `SignalRConnection`. If a host issues a command (e.g., "Mute All"), the .NET backend queries this table for all active `ConnectionIds` linked to the given `MeetingId`, ensuring targeted real-time delivery even across a multi-node Swarm.